

A Linear Weighting Scheme for Historical Searches in Recommendation Systems

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1 Introduction

This document describes a weighting scheme to incorporate historical user searches into a recommendation system. The aim is to prioritise the user's current search while allowing recent searches to influence recommendations.

2 Requirements

The weight function was designed to satisfy the following properties:

1. The function should be simple and explainable.
2. A more recent search should be weighted more than an older search.
3. The scale of dissimilarity should remain similar.

3 Combined Dissimilarity

Let

- $D_{current}$ be the dissimilarity vector of the current search,
- $D_0, D_1, \dots, D_i, \dots, D_{N-1}$ be the N vectors of ordered historical searches, most recent first,
- $W_{current} \in [0, 1]$ be the weight of the current search,
- $W(i, N, W_{current})$ be a weight function $\mathbb{N} \times \mathbb{N} \times [0, 1] \rightarrow [0, 1]$.

Then define the combined dissimilarity vector:

$$D_{combined} := W_{current} \cdot D_{current} + \sum_{i=0}^{N-1} (W(i, N, W_{current}) \cdot D_i),$$

where

$$W(i, N, W_{current}) = \frac{2(1 - W_{current})(N - i)}{N(N + 1)}.$$

Note that N and $W_{current}$ remain constant in the same search.

To allow for the case where a previous search is the only suitable recommendation, a finite penalty $\frac{1}{(i+1)}$ is added to the dissimilarity $(D_{combined})_i$ instead of an infinite penalty. This also has the property that the dissimilarity of an infinitely repeated search tends to infinity.

4 Derivation of Historical Weights

The weight function should be linearly decreasing in i , so more recent searches have more weight.

In addition, the weight function must have the property

$$W_{current} + \sum_{i=0}^{N-1} W(i, N, W_{current}) = 1,$$

keeping the scale of dissimilarity similar using a convex combination.

To satisfy both of these requirements, a weight $(N - i)$ was chosen, normalised by multiplying by the constant

$$\frac{2(1 - W_{current})}{N(N + 1)}.$$

5 Verification

Since $(N - i)$ is clearly linearly decreasing in i , it remains to show that

$$1 - W_{current} = \sum_{i=0}^{N-1} \left(\frac{2(1 - W_{current})(N - i)}{N(N + 1)} \right). \quad (1)$$

We have

$$\sum_{i=0}^{N-1} (N - i) = \sum_{k=1}^N i \quad (2)$$

$$= \frac{N(N + 1)}{2}. \quad (3)$$

Therefore,

$$\sum_{i=0}^{N-1} \left(\frac{2(1 - W_{current})(N - i)}{N(N + 1)} \right) = \frac{2(1 - W_{current})}{N(N + 1)} \sum_{i=0}^{N-1} (N - i) \quad (4)$$

$$= \left(\frac{2(1 - W_{current})}{N(N + 1)} \right) \left(\frac{N(N + 1)}{2} \right) \quad (5)$$

$$= 1 - W_{current} \quad (6)$$

as required.

6 Example

For five historical searches with

$$W_{current} = 0.8,$$

the historical weights are

$$(0.067, 0.053, 0.040, 0.027, 0.013).$$

7 Conclusion

This weighting scheme was intended as a simple modification to a k-NN recommendation system. The parameter $W_{current}$ can be tuned to individual systems.

A limitation is that if every possible item is searched infinitely, no item can be recommended. A maximum number of historical searches could be implemented in the future.